Notes towards Part III: the symmetric remainder, the dispersion form of the remainder sum, and the dimension of the weighted sieve

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October 25, 2026 — working notes, not for circulation

Abstract

These notes record three explorations around the level of distribution $1/2$ of $r(n)r(N-n)$ (Part I, Theorem 3.2; Part II, Section 7). (1) The remainder $R_d = \sum_{d_1 d_2 = d} R(d_1, d_2; N)$ that the two-dimensional sieve actually uses is, by the Atkin–Lehner identity $\sum_{u|d} G_{tu,d/u} = \prod_{p|d}(1 + W_p) G_{t,d}$, the coefficient of the projection of $G_{t,d}$ onto the joint $(+1)$-eigenspace of the involutions W_p, $p \mid d$: every Hecke component with $\eta_p(f) = -1$ for some $p \mid (d, M_1)$ cancels exactly, in particular every oldform of odd level. Numerically the cancellation is of square-root strength and no more ($|R_d|$ is 0.67–0.69 of $\sum |R(d_1, d_2)|$ for $\omega(d) = 2$ and 0.46–0.48 for $\omega(d) = 3$, against 0.707 and 0.5 for random signs); it is a $d^{o(1)}$ gain and cannot move the level. (2) The remainder sum $\sum \lambda_d \lambda'_e R(d, e; N)$ is written exactly as a trilinear form in classical Kloosterman sums $S(16N, m|h|^2; m'k)$ with the modulus in the progression $c \equiv 0 \pmod{m'}$: the character χ_{-4} disappears, no Salié sums occur, the first argument is fixed, and the uniform bound of Part I is equivalent to square-root cancellation in the sum over k. Square-root cancellation in the sum over m' would give the level $2/3$, in the sum over m as well the level 1; the Weil bound alone gives $1/4$. No unconditional level $> 1/2$ is obtained here, and the obstacles are listed. (3) The factor $2^{\Omega(n(N-n))}$ in the weights enters the final count only through a constant; what the weights $r(n)r(N-n)$ really cost is the sieve dimension, 2 instead of 1. A dimension-one sieve with the same Richert weights would give $(5, 3)$ at the level $1/2$ and $(3, 2)$ (Chen's shape $P_1 + P_2$) at the level $4/5$, against $(7, 5)$ and $(5, 3)$; the relation $r(pn') = 2r(n')$ does not lower the dimension, and the switching trick leads back to problem (2). (4) Cauchy–Schwarz over the smooth variables of the trilinear form followed by Poisson summation in $h \in \mathbb{Z}^2$ does not remove the fixed first argument: the h-box has side $\le \sqrt{mm'}$ while the modulus of the Poisson step is $\ge \sqrt{N}$, so the dual sum is longer by $N^{1/2-\theta/4}$; the diagonal of the first arrangement gives at most the level $2/5$. (5) Selberg-type mollifiers $r(n)(\sum_{u|n, u \le N^\xi} \gamma_u)^2$ keep the theta structure at the level $1/2 - 4\xi$, but a mollifier truncated below the sifting limit is identically 1 on the sifted integers and changes nothing; the density at a prime is changed only if the mollifier contains that prime, so lowering the dimension on the Richert range costs $4b$ in level, and every partial version is worse than $(7, 5)$.

Setting

We keep the notation of Parts I and II: $\chi = \chi_{-4}$, $r(n) = 4\sum_{k|n} \chi(k)$, $\Theta_{m,m'} = \theta^2(mz)\theta^2(m'z) = E_{m,m'} + G_{m,m'} \in M_2(\Gamma_0(4mm'))$ for coprime odd m, m', $a_G(N) = T(m, m'; N) - M(m, m'; N)$, W_Q the Atkin–Lehner matrices, $\eta_Q(f)$ the eigenvalues of the newforms, and $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz + d)^{-k} f(\gamma z)$, so that scalar matrices act trivially. For $N \equiv 2 \pmod 4$, $\mathcal{A} = \{n \le N : (n, N) = 1\}$, $w(n) = r(n)r(N-n)$, and for coprime square-free d_1, d_2 with $(d_1 d_2, 2N) = 1$,

$$S(d_1, d_2) = \sum_{\substack{n \in \mathcal{A} \\ d_1 | n,\ d_2 | N-n}} w(n) = X_N h(d_1) h(d_2) + R(d_1, d_2; N), \qquad R_d = \sum_{d_1 d_2 = d} R(d_1, d_2; N),$$

with $h(p) = (2p-1)/(p(p+1))$ for $p \equiv 1$, $h(q) = 1/(q(q-1))$ for $q \equiv 3 \pmod 4$ (Part I, Section 3; Part II, Section 7.2). By Part I, (16) (Part II, (4)),

$$R(d_1,d_2;N) = \sum_{t \mid N^\flat} \mu(t) \sum_{\substack{t_1 = tu \\ u \mid t}} c_t(t_1) \sum_{\substack{s_1 \mid d_1 \\ s_2 \mid d_2}} c_{d_1}(d_1 s_1) c_{d_2}(d_2 s_2) \big(a_G(N) - a_G(N/2) \big),$$

$$G = G_{t_1 d_1 s_1,\, d_2 s_2}, \quad (1)$$

where $c_p(p) = 2$, $c_p(p^2) = -1$ for split p and $c_q(q) = 0$, $c_q(q^2) = 1$ for inert q, multiplicatively; every parameter $t_1 d_1 s_1$, $d_2 s_2$ is $\equiv 1 \pmod 4$, being a product of split primes and of squares of inert primes. The scripts and outputs quoted below are in the directories `code/` and `results/` next to these notes; they import the modules of Part II.

1 The symmetric remainder R_d

1.1 Transport of a factor by an Atkin–Lehner involution

Lemma 6.1 of Part II moves the whole factor m of $\Theta_{m,m'}$ to the other side. The same computation moves any part of it.

Lemma 1.1. *Let m_1, m_2, m' be pairwise coprime odd positive integers, $L = 4m_1m_2m'$ and W_{m_1} an Atkin–Lehner matrix of level L. Then*

$$\Theta_{m_1m_2,\,m'} |_2 W_{m_1} = \chi(m_1)\, \Theta_{m_2,\,m_1m'}, \qquad \textit{hence} \qquad G_{m_1m_2,\,m'} = \chi(m_1)\, G_{m_2,\,m_1m'} |_2 W_{m_1}.$$

Proof. Write $W_{m_1} = \left(\begin{smallmatrix} m_1 a & b \\ Lc & m_1 d \end{smallmatrix}\right)$ with $m_1 ad - 4m_2m'bc = 1$; then d is odd. First, $B_{m_1m_2} W_{m_1} = m_1 B_{m_2} \gamma$ with $\gamma = \left(\begin{smallmatrix} m_1 a & b \\ 4m_2m'c & d \end{smallmatrix}\right) \in \Gamma_0(4m_2m') \subseteq \Gamma_0(4m_2)$, as one checks by multiplying out; since $\theta^2(m_2\,\cdot) \in M_1(\Gamma_0(4m_2), \chi)$,

$$\begin{aligned} \theta^2(m_1m_2\,\cdot)|_1 W_{m_1} &= (m_1m_2)^{-1/2}\, \theta^2|_1 (B_{m_1m_2} W_{m_1}) \\ &= (m_1m_2)^{-1/2} \big(\theta^2|_1 B_{m_2} \big)|_1 \gamma = m_1^{-1/2} \chi(d)\, \theta^2(m_2 z). \end{aligned}$$

Second, $W_{m_1} = \gamma' B_{m_1}$ with $\gamma' = \left(\begin{smallmatrix} a & b \\ 4m_2m'c & m_1 d \end{smallmatrix}\right) \in \Gamma_0(4m_2m') \subseteq \Gamma_0(4m')$, so

$$\theta^2(m'\cdot)|_1 W_{m_1} = \chi(m_1 d)\, \theta^2(m'\cdot)|_1 B_{m_1} = \chi(m_1 d)\, m_1^{1/2} \theta^2(m_1 m' z).$$

Multiplying, $\Theta_{m_1m_2,m'} |_2 W_{m_1} = \chi(m_1) \theta^2(m_2 z) \theta^2(m_1 m' z)$. The statement for G follows because $|_2 W_{m_1}$ preserves the orthogonal decomposition into Eisenstein and cuspidal parts (Part II, Section 6.1) and $W_{m_1}^2$ acts trivially. □

1.2 The sum over the factorisations

For odd square-free d, $s \mid d$ and $u \mid d$ put $u_s = u\,(u,s)$, so that $u_s\,(d/u)_s = ds$ and u_s is the part of ds supported on the primes of u.

Proposition 1.2. *Let d be odd and square-free, $s \mid d$, and t_1 odd with $(t_1, d) = 1$. Then*

$$\sum_{u \mid d} G_{t_1 u_s,\,(d/u)_s} = G_{t_1,\,ds} \Big|_2 \prod_{p \mid d} \big(1 + \chi(p^{v_p(ds)})\, W_{p^{v_p(ds)}} \big),$$

the W's being Atkin–Lehner matrices of level $4t_1 ds$. If every prime power $p^{v_p(ds)}$ is $\equiv 1 \pmod 4$ (which is the case for every term of (1)), the right-hand side is $2^{\omega(d)} P^+_{d,s} G_{t_1,ds}$, where $P^+_{d,s} = \prod_{p \mid d} \frac12 (1 + W_{p^{v_p(ds)}})$ is the orthogonal projection onto the joint $(+1)$-eigenspace of the commuting involutions $W_{p^{v_p(ds)}}$, $p \mid d$.

Proof. By Lemma 1.1 with $(m_1, m_2, m') = (u_s, t_1, (d/u)_s)$, $G_{t_1 u_s,(d/u)_s} = \chi(u_s) G_{t_1,ds}|_2 W_{u_s}$. For coprime exact divisors Q, Q' of the level, $W_Q W_{Q'}$ is an Atkin–Lehner matrix $W_{QQ'}$ up to a scalar and an element of $\Gamma_0(L)$ [1], so $W_{u_s} = \prod_{p|u} W_{p^{v_p(ds)}}$ on $M_2(\Gamma_0(L))$, and $\chi(u_s) = \prod_{p|u} \chi(p^{v_p(ds)})$. Expanding the product over $p \mid d$ gives the sum over $u \mid d$. The W_{p^v} are commuting self-adjoint involutions, so $\frac{1}{2}(1+W)$ is the orthogonal projection onto the $(+1)$-eigenspace and the product of the projections is the projection onto the intersection. □

Corollary 1.3. *With the notation of (1), for odd square-free d with $(d, 2N) = 1$,*

$$R_d = \sum_{t \mid N^\flat} \mu(t) \sum_{t_1 = tu,\ u \mid t} c_t(t_1) \sum_{s \mid d} c_d(ds)\, 2^{\omega(d)} \Big(a_{P^+_{d,s} G_{t_1,ds}}(N) - a_{P^+_{d,s} G_{t_1,ds}}(N/2) \Big).$$

Proof. In (1), $c_{d_1}(d_1 s_1) c_{d_2}(d_2 s_2) = c_d(ds)$ with $s = s_1 s_2$ by multiplicativity, and for fixed s the pair (s_1, s_2) is determined by (d_1, d_2): $s_1 = (s, d_1)$, $s_2 = (s, d_2)$. Hence summing (1) over $d_1 d_2 = d$ sums, for each (t, t_1, s), the coefficients of $G_{t_1 (d_1)_s,(d_2)_s}$ over all $d_1 \mid d$, and Proposition 1.2 applies. □

1.3 Which components cancel

Let now $s = 1$, $t_1 = 1$ (the terms of smallest level, which carry most of the remainder) and $n = d$. For a newform f of level $2^e M_1 \mid 4d$ let $S(f) = \mathrm{span}\{f(\delta\,\cdot) : \delta \mid 4d/(2^e M_1)\}$ and pr_f the orthogonal projection, as in Part II, Proposition 6.19.

Proposition 1.4. *Let d be odd and square-free.*

(i) *If $p \mid M_1$, W_p acts on $S(f)$ as the scalar $\eta_p(f)$. If $p \mid d$, $p \nmid M_1$, W_p maps $f(\delta\,\cdot)$ to $(\delta'/\delta) f(\delta'\cdot)$ with $\delta' = \delta p$ or δ/p (Part II, Lemma 6.8), so that $S(f)$ splits into two W_p-eigenspaces of equal dimension.*

(ii) *$W_d = \prod_{p|d} W_p$ acts on $G_{1,d}$ as $\chi(d)$ (Part II, Remark 6.9(i)).*

(iii) *Consequently $P^+_{d,1} \mathrm{pr}_f G_{1,d} = 0$ whenever $\eta_p(f) = -1$ for some $p \mid M_1$. In particular, writing $\varepsilon(f) = -\eta_{2^e}(f)\eta_{M_1}(f)$ for the root number and using $\eta_4 = -1$ for $e = 2$ (Part II, Proposition 6.17) and Theorem 6.7(ii) of Part II (the component $\mathrm{pr}_f G_{1,d}$ is non-zero only if $L(1,f)L(1,f\otimes\chi) \neq 0$, hence only if $\varepsilon(f) = +1$):*

- *$e = 0$, $M_1 > 1$: $\varepsilon(f) = +1$ forces $\eta_{M_1}(f) = -1$, so some $\eta_p(f) = -1$ and the whole component cancels;* every oldform of odd level cancels in R_d*;*
- *$e = 1$: $\varepsilon(f) = +1$ forces $\eta_{M_1}(f) = -\eta_2(f)$; the component survives only if $\eta_2(f) = -1$, i.e. $a_f(2) = +1$, and $\eta_p(f) = +1$ for every $p \mid M_1$;*
- *$e = 2$: $\varepsilon(f) = \eta_{M_1}(f)$, so $\eta_{M_1}(f) = +1$ is automatic, and the component survives iff $\eta_p(f) = +1$ for every $p \mid M_1$; for the newforms of level $4d$ this is the condition $\eta_p = +1$ for all $p \mid d$.*

What survives of a surviving component is its part in $\bigcap_{p|d,\, p\nmid M_1} S(f)^{+,p}$, and it enters R_d with the factor $2^{\omega(d)}$.

Proof. (i) is Lemma 6.8 of Part II with $Q = p$; (ii) is Lemma 1.1 with $(m_1, m_2, m') = (d, 1, 1)$ and $\Theta_{d,1} = \Theta_{1,d}$. For (iii), $P^+_{d,1}$ contains the factor $\frac{1}{2}(1 + W_p)$, which annihilates $S(f)$ when $W_p = -1$ on $S(f)$. The sign conditions are the functional equation $\Lambda(s, f) = \varepsilon(f)\Lambda(2 - s, f)$ with $\varepsilon(f) = -\eta_{\mathrm{level}}(f)$ in weight 2. □

Remark 1.5. (i) For $d = p$ prime nothing cancels: $W_p = \chi(p)$ on $G_{1,p}$, and indeed $R(p,1) = R(1,p)$ by the symmetry $n \leftrightarrow N - n$ of w. The claim that "all f with $\eta_p(f) = -1$ for some $p \mid (d, M_1)$ cancel" is correct, but for $d = p$ the set of such f in $G_{1,p}$ is empty. The cancellation is between the $2^{\omega(d)-1}$ *unordered* splittings $\{d_1, d_2\}$; for $d = p_1p_2$ it is between $R(d,1)$ and $R(p_1,p_2)$, i.e. between $G_{1,d}$ and $G_{p_1,p_2} = G_{1,d}|W_{p_1}$, and it removes the $(-,-)$-eigenspace, which by (iii) contains all the oldforms of odd level, the oldforms of level $2p_i$ with $a_f(2) = -1$, and half (in the sense of (i)) of every other component, the newforms of level $4d$ with $\eta_{p_1} = \eta_{p_2} = -1$ included.

(ii) The identity of Proposition 1.2 holds for the terms with $s > 1$ (parameters with squares of primes) as well; there W_{p^2} acts on $S(f)$ with both eigenvalues even when $p \mid M_1$, and the description (iii) does not apply verbatim. These terms are smaller by $(ds/d)^{-1/2}$ in mean square.

(iii) The corresponding statement for the sum with the weight $\chi(m)$ over all factorisations $m \mid n$ is Proposition 6.19 of Part II; here $\chi(m) = 1$ for every m that occurs, and the point is that the sieve itself performs this sum.

1.4 Numerical results

The script `symmetric_remainder.py` computes every $R(d_1, d_2; N)$ exactly (direct summation of $w(n)$ over the progression, main term $X_N h(d_1)h(d_2)$ from Part I, (17)–(18)) for all admissible pairs with $d_1d_2 \le N^{3/4}$ and 24 values of $N \equiv 2 \pmod 4$ near $2 \cdot 10^6$ and $3 \cdot 10^6$, and compares $A_{\mathrm{sym}} = \sum_{d \le D} |R_d|$ with $A_{\mathrm{abs}} = \sum_{d_1d_2 \le D} |R(d_1,d_2)|$ and with the random-sign baseline A_{rnd} (independent signs on the unordered splittings, averaged over 40 draws). The symmetry $R(d_1,d_2) = R(d_2,d_1)$ holds to 10^{-10}. Table 1 gives the averages over the 24 values of N (`results/symmetric_remainder.txt`).

θ	$A_{\mathrm{sym}}/A_{\mathrm{abs}}$			$A_{\mathrm{sym}}/A_{\mathrm{rnd}}$			with weight $4^{\omega(d)}$	
	all d	$\omega(d)=2$	$\omega(d)=3$	all d	$\omega(d)=2$	$\omega(d)=3$	all d	random
0.50	0.750	0.669	0.458	0.968	0.952	0.898	0.602	
0.60	0.708	0.687	0.462	0.975	0.975	0.924	0.547	
0.70	0.672	0.688	0.477	0.976	0.977	0.949	0.503	
0.75	0.659	0.692	0.484	0.980	0.982	0.962	0.487	
random signs		0.707	0.500	1	1	1		

Table 1: $\sum_{d \le N^\theta} |R_d|$ against $\sum_{d_1d_2 \le N^\theta} |R(d_1,d_2;N)|$, averages over 24 values of N; the standard deviations over N are 0.01–0.05 (0.13 for $\omega = 3$ at $\theta = 0.5$, where there are only 16 such moduli). For $\omega(d) = 1$ the ratio is 1 exactly.

The cancellation is real, exact in the sense of Proposition 1.4, and of square-root strength: $|R_d|$ is 0.67–0.69 of $\sum |R(d_1,d_2)|$ for two prime factors (random signs: $1/\sqrt{2} = 0.707$) and 0.46–0.48 for three (random: $1/2$), i.e. 2–5% below the random-sign value. The spectral reason is that the norm of $G_{1,d}$ is close to equidistributed over the joint eigenspaces. The script `symmetric_norm_share.py` computes, from the exact coefficients of all $G_{u,d/u}$ up to 2^{18}, the Rankin–Selberg shares $\|P_\varepsilon G_{1,d}\|^2/\|G_{1,d}\|^2$ of every sign pattern ε (`results/symmetric_norm_share.txt`); the patterns with $\prod \varepsilon_p \ne \chi(d)$ carry $< 10^{-34}$, confirming (ii) numerically, and for $d = p_1p_2$ with both primes split the share of $(+,+)$ is 0.50–0.62 (0.503 for $221 = 13 \cdot 17$, 0.625 for $445 = 5 \cdot 89$), for $\omega(d) = 3$ the share of $(+,+,+)$ is 0.35–0.38 and for $\omega(d) = 4$ it is 0.19–0.20. These are shares of the norm; the sieve sees the coefficient $a(N) - a(N/2)$ with $N \equiv 2 \pmod 4$, and at even N the newforms of level $4d$ contribute nothing ($a_\varphi(2) = 0$), so the sieve weighs the old and the new parts differently. Restricted to the combination $a(2n') - a(n')$, $n' \equiv 1 \pmod 4$, $(n', d) = 1$, the same script gives the share 0.39–0.47 for $(+,+)$ when $\omega(d) = 2$, 0.20 for $(+,+,+)$ and 0.105–0.109 for $(+,+,+,+)$, against 1/2, 1/4, 1/8 for equidistribution; and the remainders themselves give,

as the empirical share $\langle (R(d,1)+R(p_1,p_2))^2\rangle / \langle 2(R(d,1)^2+R(p_1,p_2)^2)\rangle$ over the 24 values of N and 67 moduli $d = p_1p_2 \le 2000$ with both primes split, the value 0.424. The two determinations agree, and both are below $1/2$ by the amount that Proposition 1.4(iii) removes (the oldforms of odd level, which are in the $(-,-)$-eigenspace).

1.5 Consequence for the sieve

Corollary 1.3 replaces, in the remainder sum $2\sum_{m\mid P(z),\,m<y} 4^{\omega(m)}|R_m|$ of the two-dimensional sieve (Part II, (20)), the bound $|R_m| \le \sum_{d_1d_2=m}|R(d_1,d_2)|$ by $2^{\omega(m)}\sum_{t,t_1,s}|c\,a_{P^+G}(N')|$ with $\|P^+G\| \le \|G\|$, and the numerics say that in mean square the gain is the factor $(\text{share})^{1/2} \approx 2^{-(\omega(m)-1)/2}$ against the trivial $2^{\omega(m)-1}$ terms: the algebraic cancellation realises the square-root cancellation among the $2^{\omega(m)-1}$ terms that one would expect from independent signs, and essentially nothing beyond it, because the eigenspaces share the norm almost equally. Since $2^{\omega(m)} = m^{o(1)}$, this is absorbed by the factor N^η of condition (F1) in the proof of Part I, Theorem 3.2, and does not change the level of distribution; it changes neither $(7,5)$ nor the constant 0.25 of Part II, Theorem 7.1 (the constant is governed by the main term, not by the remainder). The exact identity is nevertheless worth recording: it says that the two-dimensional sieve for $n(N-n)$ sees, at each modulus d, only the W_p-invariant part of $G_{t,d}$, and therefore none of the oldforms of odd level.

2 The dispersion form of the remainder sum

2.1 The circle modulo c

Lemma 2.1. *Let c be odd and square-free, $(a,c)=1$ and $h=(h_1,h_2)\in\mathbb{Z}^2$. Then*

$$F_c(a;h) := \sum_{\substack{x,y \bmod c\\ x^2+y^2\equiv a\ (c)}} e\Big(\frac{h_1x+h_2y}{c}\Big) = \sum_{g\mid(c,h_1,h_2)} g\,\chi\Big(\frac{c}{g}\Big)\,S\Big(a,\ \overline{4}\,\frac{h_1^2+h_2^2}{g^2};\ \frac{c}{g}\Big),$$

with $S(m,n;c) = \sum^*_{t \bmod c} e((mt+n\bar t)/c)$ the classical Kloosterman sum ($S(\cdot;1)=1$). For prime c this reads $F_c(a;h) = \chi(c)S(a,\overline{4}|h|^2;c) + c\,[h\equiv 0\ (c)]$.

Proof. Detect the congruence by $\frac{1}{c}\sum_{t \bmod c} e(t(x^2+y^2-a)/c)$. For $(t,c)=g$, $t=gt'$, the sum $\sum_{x \bmod c} e((tx^2+h_1x)/c)$ vanishes unless $g\mid h_1$, and then equals $g\,e(-\overline{4t'}(h_1/g)^2/(c/g))\,G(t';c/g)$ by completing the square, where $G(t';c') = \sum_x e(t'x^2/c') = (t'/c')\,\varepsilon_{c'}\sqrt{c'}$ with $\varepsilon_{c'}^2 = \chi(c')$ for odd square-free c'. Hence $G(t';c')^2 = \chi(c')c'$, and

$$F_c(a;h) = \sum_{g\mid(c,h)} \frac{g^2}{c}\cdot\chi\Big(\frac{c}{g}\Big)\frac{c}{g}\sideset{}{^*}\sum_{t' \bmod c/g} e\Big(\frac{-t'a-\overline{4t'}|h/g|^2}{c/g}\Big) = \sum_{g\mid(c,h)} g\,\chi\Big(\frac{c}{g}\Big)S\Big(a,\overline{4}|h/g|^2;\frac{c}{g}\Big),$$

using $S(-m,-n;c') = S(m,n;c')$. The script `kloosterman_check.py` verifies the identity for all a, h and all odd primes $c\le 61$, and the composite form for nine square-free composite $c\le 105$ (`results/kloosterman_check.txt`). □

So the Fourier transform of the circle $x^2+y^2\equiv a$ modulo c is a *classical* Kloosterman sum whose second argument is the norm $|h|^2$ of the dual variable, twisted only by the sign $\chi(c)$; no Salié sum and no character sum of a different type occurs. This is the arithmetic reason why $r(n)$ in arithmetic progressions behaves like the divisor function: the Weil bound gives the asymptotic for $\sum_{n\equiv a\,(c),\,n\le Y} r(n)$ for $c\le Y^{2/3-\varepsilon}$, exactly as for $\tau(n)$.

2.2 The remainder as a trilinear form in Kloosterman sums

Let $m \equiv m' \equiv 1 \pmod 4$ be coprime, $(mm', N) = 1$, $N \equiv 2 \pmod 4$, and let $T^{\mathrm{odd}}(m, m'; N)$ be the part of $T(m, m'; N) = \sum_{m\alpha + m'\beta = N} r(\alpha) r(\beta)$ with α odd, so that $a_G(N) - a_G(N/2) = T^{\mathrm{odd}} - M^{\mathrm{odd}}$ in (1). In T^{odd} every term with $r(\alpha)r(\beta) \neq 0$ has $\alpha \equiv \beta \equiv 1 \pmod 4$, so $x^2 + y^2 = \alpha$ has $x + y$ odd, and $r(\beta) = 8 \sum_{k \mid \beta,\, k < \sqrt{\beta}} \chi(k) + 4\chi(\sqrt{\beta})[\beta = \square]$ by the symmetry $k \leftrightarrow \beta/k$ ($\chi(\beta/k) = \chi(k)$ for $\beta \equiv 1 \pmod 4$). Writing $\beta = (N - m\alpha)/m'$ and noting that $(k, m) = 1$ automatically (a common prime would divide N),

$$T^{\mathrm{odd}}(m, m'; N) = 8 \sum_{k < \sqrt{N/m'}} \chi(k)\, \mathcal{N}_k + 4 \sum_{j \geq 1} \chi(j)\, r^{\mathrm{odd}}\Big(\frac{N - m'j^2}{m}\Big), \tag{2}$$

$$\mathcal{N}_k = \#\big\{(x, y) \in \mathbb{Z}^2 :\ x + y \text{ odd},\ x^2 + y^2 \equiv N\bar{m} \ (m'k),\ 0 < x^2 + y^2 < Y_k := (N - m'k^2)/m\big\},$$

where $r^{\mathrm{odd}}(\alpha) = r(\alpha)[\alpha \text{ odd}]$ and the second sum runs over j with $m \mid N - m'j^2$; it is the number of representations of N by the ternary form $m(x^2 + y^2) + m'j^2$ with a sign, and is $O(N^{1/2+\varepsilon} m'^{-1/2})$ trivially.

For the lattice-point count $\mathcal{N}_k$ (with a smooth cutoff w at Y_k, the usual sharp-to-smooth passage costing a boundary term), Poisson summation on the residue classes modulo $2c$, $c = m'k$, gives

$$\mathcal{N}_k = \frac{Y_k}{2c^2} \sum_{\substack{h \in \mathbb{Z}^2 \\ h_1 \equiv h_2 \ (2)}} \hat{w}\big(h\sqrt{Y_k}/(2c)\big)\, F_c\big(N\bar{m};\ \bar{2}h\big), \tag{3}$$

the factor $2[h_1 \equiv h_2]$ (absorbed in $1/(2c^2) = 2/(2c)^2$) being the Fourier transform of the parity condition and $\bar{2}h$ the Chinese-remainder twist. The term $h = 0$ is $\frac{Y_k}{2c^2}\rho(N\bar{m}; c)$ with $\rho(a; c) = \#\{x^2 + y^2 \equiv a\ (c)\} = c\prod_{p \mid c}(1 - \chi(p)/p)$, and the sum over k of these terms is the Siegel main term $M^{\mathrm{odd}}(m, m'; N)$ (the singular series of the quaternary form, written through the divisor structure of $r(\beta)$). By Lemma 2.1, for the leading term $g = 1$, $\chi(k) \cdot \chi(m'k) = \chi(m') = 1$ and

$$S\big(N\bar{m},\ \overline{16}\,|h|^2;\ m'k\big) = S\big(16N,\ m|h|^2;\ m'k\big).$$

The character $\chi(k)$ of the divisor expansion is exactly cancelled by the sign $\chi(c)$ of the Gauss sums. We have proved:

Proposition 2.2. *With the notation above,*

$$a_G(N) - a_G(N/2) = 8 \sum_{k < \sqrt{N/m'}} \frac{Y_k}{2(m'k)^2} \sum_{\substack{h \neq 0 \\ h_1 \equiv h_2 \ (2)}} \hat{w}\Big(\frac{h\sqrt{Y_k}}{2m'k}\Big) \times \sum_{g \mid (m'k, h)} g\, \chi\Big(\frac{m'k}{g}\Big)\chi(k)\, S\Big(N\bar{m}, \overline{16}\,\Big|\frac{h}{g}\Big|^2; \frac{m'k}{g}\Big) + \Sigma_\square + \Sigma_\partial,$$

where $\Sigma_\square$ is the ternary term of (2) and Σ_∂ the smoothing error; the term $g = 1$ is

$$8 \sum_{k < \sqrt{N/m'}} \frac{Y_k}{2(m'k)^2} \sum_{\substack{h \neq 0 \\ h_1 \equiv h_2 \ (2)}} \hat{w}\Big(\frac{h\sqrt{Y_k}}{2m'k}\Big) S\big(16N,\ m|h|^2;\ m'k\big). \tag{4}$$

Consequently the remainder sum of the sieve at the level N^θ, after the decomposition (1) (with the sieve weights and the coefficients c collected into $\Lambda_m, \Lambda'_{m'} \ll 3^\omega$, and with $m \leq m'$ by the

symmetry $n \leftrightarrow N-n$), is a trilinear form

$$\mathfrak{B} = \sum_{m'} \Lambda'_{m'} \sum_{c \equiv 0 \ (m')} \sum_{m} \Lambda_m \sum_{j} b_{m,j}(c)\, S(16N,\ mj;\ c),$$

$$b_{m,j}(c) = \frac{4Y_{c/m'}}{c^2} \sum_{|h|^2=j} \hat{w}\Big(\frac{h\sqrt{Y_{c/m'}}}{2c}\Big), \quad (5)$$

with the modulus $c = m'k$ in the progression $c \equiv 0 \pmod{m'}$, $c \le \sqrt{Nm'}$, the first argument $16N$ fixed, the second argument $n = mj$ with j a norm, and smooth weights in c and h.

Remark 2.3. (i) The dual variable satisfies $|h| \ll m'k\sqrt{m/N}\,N^{\varepsilon}$; so the inner sum of (4) is empty unless $k \gg \sqrt{N/m}/m'$, the modulus runs in $\sqrt{N/m} \ll c \le \sqrt{Nm'}$, and $n = m|h|^2 \ll m^2m'^2k^2/N \le m^2m'$. The sum over k has length $K \asymp \sqrt{N/m'}$.

(ii) Opening the Kloosterman sum, (4) is the Kloosterman refinement of the circle method for the quaternary form $Q_{m,m'}$, with the moduli $m'k$ ordered by the divisor structure of $r(\beta)$. Summing (4) over k with the Kuznetsov formula for $\Gamma_0(m')$ expresses it, for each (m,j), through the Fourier coefficients $\rho_\ell(16N)\overline{\rho_\ell(mj)}$ of the Maass forms (and the Eisenstein series and the holomorphic forms) of level m'; the further sum $\sum_j r(j)\rho_\ell(mj)j^{-s}$ is, formally, the Rankin–Selberg convolution of u_ℓ with θ^2, i.e. $L(s,u_\ell)L(s,u_\ell\otimes\chi)$ up to the factors at $p \mid m$, the Maass analogue of Theorem 6.7 of Part II. The spectral side of the dispersion method is thus the family of products $L(\frac12+it_\ell,u_\ell)L(\frac12+it_\ell,u_\ell\otimes\chi)$ over the Maass spectrum of the *sieve modulus* m', weighted by $\rho_\ell(16N)$, while the holomorphic method of Parts I–II works at the level $4mm'$.

(iii) This is the two-sided version of Linnik's problem $N = p + x^2 + y^2$, for which the dispersion method was invented: there $\sum_p r(N-p) = 4\sum_p \sum_{k \mid N-p} \chi(k)$ is a Titchmarsh-type sum with the moduli k up to N and the primes as the rough variable. Here both sides carry r, both carry sieve weights, and the divisor structure of one side (β) provides the moduli $m'k$ while the lattice structure of the other side ($\alpha = x^2+y^2$) provides the Kloosterman sums.

2.3 Exponent bookkeeping

In (4), for each k the number of h is $\asymp (m'k)^2m/N$ and the weight $Y_k/(2(m'k)^2) \asymp N/(m(m'k)^2)$, so each k contributes $O(1)$ terms of weight $O(1)$ on average.

- *Weil bound only.* $|S(16N, m|h|^2; c)| \le d(c)c^{1/2}(|h|^2,c)^{1/2}$ gives $|a_G(N) - a_G(N/2)| \ll N^{\varepsilon}\sum_{k \le \sqrt{N/m'}}(m'k)^{1/2} \ll N^{3/4+\varepsilon}m'^{-1/4}$, and the sum of absolute values over $mm' \le N^{\theta}$ is $\ll N^{3/4+\theta+\varepsilon}$: level $1/4$.

- *The theta-function bound.* Theorem 1.2(ii) of Part I, $|a_G(N)| \ll N^{1/2+\varepsilon}$, is smaller than the Weil bound by exactly $(N/m')^{1/4} = K^{1/2}$: it is equivalent to square-root cancellation in the sum over the moduli $c = m'k \equiv 0 \pmod{m'}$, $k \le K$, for the specific sequence $n = m|h|^2$ of second arguments. Summing absolute values over $mm' \le N^{\theta}$ gives the level $1/2$ (Part I, Theorem 3.2).

- *Cancellation over m'.* If $\sum_{m' \sim E} \Lambda'_{m'}\big(a_{G_{m,m'}}(N) - a_{G_{m,m'}}(N/2)\big) \ll N^{1/2+\varepsilon}E^{1/2}$ for each $m \le \Delta$ (square-root cancellation among the E values of m'), the remainder is $\ll N^{1/2+\varepsilon}\Delta E^{1/2}$, which is $\ll N^{1-\delta}$ for $\Delta^2E \le N^{1-3\delta}$; with the balanced split $\Delta = E = N^{\theta/2}$ (the hardest case, which the sieve needs) this is $\theta < 2/3$. The same conclusion follows from Part II, Proposition 4.2, with a saving $E^{-1/2}$ in place of $(de)^{-\eta}$.

- *Cancellation over m' and m.* Square-root cancellation over both gives $N^{1/2+\varepsilon}(\Delta E)^{1/2}$ and the level $1-\varepsilon$.

The data of Part II, Section 8.2 (the signed sums over (d,e) are of the size of the root mean square, for all three weightings and all $\theta \le 3/4$) are consistent with the last two items; they say nothing about a proof.

2.4 What the dispersion method must deliver, and what is available

The form (5) is the type of sum treated by Deshouillers and Iwaniec [5, Theorems 9–12], bilinear forms $\sum_{c\equiv 0\,(q)}\sum_m\sum_n a_m b_n g(c,m,n)S(m,\pm n;c)$ bounded through the Kuznetsov formula and the spectral large sieve, and by Drappeau's extension [6, Theorem 2.1] with a congruence condition on the modulus: in his normalisation, for $b_{n,r,s}$ supported on $n\le N_0$, $r\sim R$, $s\sim S$, moduli sc with $c\sim C$ in a class modulo q and $d\sim D$ in a class modulo q,

$$\sum\sum b_{n,r,s}\, g(c,d,n,r,s)\, e\Big(\frac{n\,\overline{rd}}{sc}\Big) \ll (qCDN_0RS)^{\varepsilon}\, q^{3/2} K\, \|b\|_2,$$
$$K^2 = qCS(RS+N_0)(C+RD) + C^2DS\sqrt{(RS+N_0)R} + D^2N_0R/S.$$

To apply such a bound to (5) one has to overcome three specific features.

(a) *The first argument is fixed.* In (5) there is no average over the first argument $16N$: the large-sieve inequalities for the Fourier coefficients $\rho_\ell(16N)$ at a single $16N \asymp N \gg c$ cost the factor $N^{1/2}$-type terms of the spectral large sieve (and, through the exceptional spectrum, N^{ϑ} with Kim–Sarnak $\vartheta = 7/64$), whereas the known applications (BFI, Titchmarsh divisor problem) always average over a long variable in both arguments. The one-sided average that is available is over $n = mj$ with the coefficient $b_n = \sum_{mj=n}\Lambda_m r(j)\cdot(\text{weights})$, of length $\ll \Delta^2 E$.

(b) *The sieve weight sits on the divisibility class of the modulus.* The sum over m' in (5) is over the *class* $c\equiv 0 \pmod{m'}$ with the weight $\Lambda'_{m'}$. This is where BFI use the well-factorable structure of the linear sieve weights, $\Lambda' = \Lambda'_1 * \Lambda'_2$ with $m' = m'_1m'_2$ in prescribed ranges, so that the modulus $c = m'_1m'_2k$ has two smooth factors; the weights of the two-dimensional sieve (DHR) are not well-factorable in Iwaniec's sense, and one would first have to go back to a vector or β-sieve with well-factorable weights, at the price of $(7,5)$.

(c) *The hard range.* The Kloosterman regime is $c\in[\sqrt{N/m},\sqrt{Nm'}]$, always beyond $\sqrt{N}$; the Bessel argument $4\pi\sqrt{16N\cdot mj}/c \asymp m$ is of constant size in k, so the Kuznetsov transform does not localise the spectrum, and the whole spectrum of level m' up to $t_\ell \ll m$ contributes.

We have not carried out the estimation, and we make no claim of a level $> 1/2$. What the computation establishes is the exact shape of the problem: after the theta-function saving (the sum over k), the remainder sum is $N^{1/2+\varepsilon}\sum_{m\le m'}|\Lambda_m\Lambda'_{m'}|$, and any power saving in the sum over the sieve modulus m' of the Kloosterman sums (4), uniform in $m\le m'$, raises the level beyond $1/2$ (Part II, Proposition 4.2); the heuristic ceiling of the method, square-root cancellation over m', is the level $2/3$, which would give $(6,4)$ for every N (Part II, Table 4). Whether the specific levels $4/7$ or $13/25$ are attainable is open.

3 The dimension of the weighted sieve

3.1 Where 2^Ω enters

The weighted sieve of Part II, Section 7.2, gives $W \ge X_N V(z)(L_2 - \lambda R_2 - o(1))$ for

$$W = \sum_{\substack{n\in\mathcal{A}\\(n(N-n),P(z))=1}} w(n)\Big(1-\lambda\sum_{\substack{z\le p<y\\ p\mid n(N-n)}}\Big(1-\frac{\log p}{\log y}\Big)\Big),$$

and the count of integers follows from $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \le 2^{k+4}$ on the integers with positive weight. The pairs (k, k') are decided by the inequalities $1/\lambda + 2/b \le k+1$ and $1/\lambda + 1/b \le k'+1$ together with $L_2 - \lambda R_2 > 0$; the factor 2^Ω enters only in the final division by $\max w$, i.e. in the constant 0.25 of Theorem 7.1 and nowhere else. Removing it would improve the constant by at most 2^{k+4} and would not change (k, k'). The Buchstab identity does not help here either: W is a sum of non-negative weights $w(n) \cdot$ (Richert weight), and writing $w(n) = 2^j \cdot 16$ on $\{\Omega(n(N-n)) = j\}$ one has $\#\{n\} \ge \sum_j W_j \ge W/2^{k+4}$ with $W_j \ge 0$; without a lower bound for the individual W_j, which the sieve does not provide, the division by 2^{k+4} is the best possible.

3.2 The dimension

What the weights $r(n)r(N-n)$ really cost is the dimension. Under w, the density of $p \mid n(N-n)$ on $\mathcal{A}$ is $2h(p) \sim 4/p$ for split p and $2h(q) \sim 2/q^2$ for inert q, so $\sum_{p<z} \omega(p)/p = 2\log\log z + O(1)$: dimension $\kappa = 2$. Under the indicator weight $b(n)b(N-n)$ (n and $N-n$ sums of two squares) the density of $p \mid n$ among the sums of two squares is $\sim 1/p$ for split p (since $n = pn'$ is a sum of two squares iff n' is) and $\sim 1/q^2$ for inert q, so the two sides together have $\sum_{p<z} \omega(p)/p = \log\log z + O(1)$: dimension $\kappa = 1$, the dimension of Goldbach's problem. The factor 2 in $r(pn') = 2r(n')$ is this doubling of the density, not a weight attached to the sifted integers that could be dropped.

The script `dimension_benchmark.py` computes, with the sieve functions of Part II (`dhr_sieve.py`, $\kappa = 1$ giving the linear sieve functions), the smallest attainable thresholds

$$\min\Big(\frac{R}{L} + \frac{2}{b}\Big), \qquad \min\Big(\frac{R}{L} + \frac{1}{b}\Big), \qquad L = f_\kappa\Big(\frac{\theta}{a}\Big), \quad R = \kappa \int_a^b \Big(1 - \frac{t}{b}\Big) F_\kappa\Big(\frac{\theta - t}{a}\Big) \frac{\mathrm{d}t}{t},$$

over $0 < a < b < \theta$, $a < \theta/\beta_\kappa$, for both dimensions (`results/dimension_benchmark.txt`):

θ	$\kappa = 2$ (weights $r(n)r(N-n)$)			$\kappa = 1$ (indicator weights, hypothetical)		
	joint	side	(k, k')	joint	side	(k, k')
1/2	7.561	5.065	(7, 5)	5.181	3.144	(5, 3)
2/3	6.373	4.313	(6, 4)	4.171	2.602	(4, 2)
4/5	5.741	3.889	(5, 3)	3.663	2.313	(3, 2)
19/20	5.212	3.519	(5, 3)	3.255	2.069	(3, 2)
1	5.065	3.414	(5, 3)	3.144	2.000	(3, 2)

The $\kappa = 2$ column reproduces Part II, Table 4, and at $\theta = 1$ Wang's $P_2 + P_3$ for the unweighted $n(N-n)$. The $\kappa = 1$ column is what the same Richert weights would give for the sequence $b(n)b(N-n)$ if its remainders had the level θ: $(5, 3)$, i.e. $P_2 + P_3$, already at $\theta = 1/2$, and $(3, 2)$, i.e. $N = P_1 + P_2$, Chen's shape, from $\theta = 4/5$ on. The dimension costs two prime factors on each count at every level. (The two thresholds are attained at different (a, b); a simultaneous pair may need one more unit on one side, as in Part II, Table 4.)

3.3 Why $r(pn') = 2r(n')$ does not lower the dimension

Extracting a split prime $p \mid n$ from $S(\mathcal{A}_p, z)$, $n = pn'$ with $p \nmid n'$, gives $w(n) = 2r(n')r(N - pn')$, and the sum over n' is the sifted sum of the sequence $\{n' \le N/p\}$ with the weights $r(n')r(N - pn')$, whose congruence sums $\sum_{d|n',\, e|N-pn'} r(n')r(N - pn')$ are the $T(t_1\tilde{m}p, \tilde{m}'; N)$ of the pair (dp, e). Its density at a split prime q is again $2h(q)$ and its dimension again 2; the factor 2 is the main term $2h(p)X_N$ of $S(\mathcal{A}_p, z)$ itself. Selberg's Λ^2 weights give an upper bound for $S(\mathcal{A}_p, z)$ with the same main term and the same remainders. Chen's switching trick would bound $\sum_{p \sim P} S(\mathcal{A}_p, z)$ by sieving the sequence $\{pn'\}$ jointly in p and n', which requires the remainders $\sum_{p \sim P} \big(T(dp, e; N) - M(dp, e; N)\big)$ for $de \le N^\theta$ and $P \ge y = N^b$ with

$b \approx 0.4$: moduli up to $N^{\theta+b} > N^{1/2}$, i.e. cancellation in the sum over the prime variable p of the coefficients $a_{G_{dp,e}}(N)$. That is the problem of Section 2 with the modulus $m = dp$ carrying a prime variable, not an independent route.

3.4 What would lower it

A level of distribution for mixed sequences $b(n)r(N-n)$ or $b(n)b(N-n)$ (dimension $3/2$ and 1) would give the $\kappa = 1$ column, or an intermediate one, with the same sieve; but $\sum_{d|n} b(n)e(nz)$ is not a modular form, and the only tools for b are the sieve itself (the Brun–Hooley type results for sums of two squares in short intervals and progressions) or, for $1_{\text{prime}}(n)r(N-n)$, the dispersion method of Linnik and Hooley. We record the comparison as the quantitative content of "breaking the 2^{Ω} penalty": it is the passage from $(7,5)$ to $(5,3)$ at the level $1/2$, and from $(5,3)$ to $(3,2)$ at the level 1, and it requires a new source of remainder control, not a rearrangement of the weights.

4 Cauchy–Schwarz and Poisson against the fixed first argument

The obstacle (a) of Section 2.4 is that the first argument $16N$ of the Kloosterman sums in (5) is not averaged. Here we examine whether it can be dualised away: fix $m \asymp \Delta$, apply Cauchy–Schwarz over the smooth variables, expand the square over $m_1', m_2' \asymp E$, and apply Poisson summation to $h \in \mathbb{Z}^2$. The outcome is negative, for a reason that is worth stating precisely: in (5) the box of h has side $H \le \sqrt{mm'}$, while the modulus of the Poisson step is at least $\sqrt{N}$; the dual of a short sum is a long sum.

4.1 The sizes

Fix $m \asymp \Delta$ and dyadic $m' \asymp E$, $k \asymp K \le \sqrt{N/E}$, and put $c = m'k$. In (4) the weight is $\asymp W_0 := N/(mE^2K^2)$ and $\hat{w}$ localises h to $|h| \le H := EK\sqrt{m/N}$, so that

$$H^2 = \frac{E^2K^2m}{N} \le mE = mm', \qquad H^2W_0 \asymp 1, \qquad \#\{(k,h)\} \asymp KH^2. \tag{6}$$

The Kloosterman regime is $K \ge \sqrt{N/m}/E$ (Remark 2.3(i)), and the top range $K \asymp \sqrt{N/E}$, where $H^2 \asymp mm'$, dominates.

4.2 Two Cauchy–Schwarz arrangements

Let $\mathfrak{B}_m = \sum_{m'} \Lambda'_{m'} \sum_k \sum_h w\, S(16N, m|h|^2; m'k)$ be the m-th piece of (5); the target is $|\mathfrak{B}_m| \le N^{1-\delta}/\Delta$, and the theta-function bound gives $|\mathfrak{B}_m| \ll EN^{1/2+\varepsilon}$.

(A1) Smooth variables outside, m' inside. By Cauchy–Schwarz over (k,h),

$$|\mathfrak{B}_m|^2 \le \Big(\sum_{k,h} W_0^2\Big) \sum_{k,h} \Big|\sum_{m'} \Lambda'_{m'} S(16N, m|h|^2; m'k)\Big|^2$$
$$\asymp \frac{N}{mE^2K} \Big(\sum_{m'} \sum_{k,h} |S(16N, m|h|^2; m'k)|^2 + \text{off-diagonal}\Big).$$

The diagonal is the mean square of Kloosterman sums over h, which is $\asymp H^2c$ by Weil (and is sharp: the mean of $|S(a,n;c)|^2$ over n is $\varphi(c)$), so it contributes $E \cdot K \cdot H^2 \cdot EK = E^2K^2H^2$ and

$$|\mathfrak{B}_m|^2 \ll \frac{N}{mE^2K} E^2K^2H^2 = E^2K^3, \qquad |\mathfrak{B}_m| \ll EK^{3/2} = E^{1/4}N^{3/4} \quad (K = \sqrt{N/E}),$$

even if the off-diagonal terms vanished. This beats $EN^{1/2}$ only for $E > N^{1/3}$, and the sieve condition $\Delta E^{1/4}N^{3/4} \le N^{1-\delta}$ with the balanced split $\Delta = E$ gives $E \le N^{1/5}$, i.e. the level $2/5 < 1/2$. The diagonal already loses, because Cauchy–Schwarz replaces the theta-function saving $K^{1/2}$ on the sum over k by the Weil bound: the smooth variables are too short to carry the dispersion.

(A2) h outside, (m',k) inside. Now $|\mathfrak{B}_m|^2 \le (\sum_h W_0^2)\sum_h |\sum_{m',k} S|^2$, the diagonal is $H^2 \cdot EK \cdot EK$, and with (6)

$$|\mathfrak{B}_m|^2 \ll H^2 W_0^2 \cdot H^2E^2K^2 = E^2K^2, \qquad |\mathfrak{B}_m| \ll EK = \sqrt{EN},$$

which is the level-2/3 target of Section 2.3; but this diagonal *is* the statement that the sum over the moduli $c = m'k$ has square-root cancellation, so (A2) is a reformulation of the problem, not a method, and everything is in the off-diagonal terms

$$C(c_1,c_2) = \sum_h W\Big(\frac{h}{H}\Big) S(16N, m|h|^2; c_1)\, S(16N, m|h|^2; c_2), \qquad c_i = m_i'k_i,\ c_1 \ne c_2. \tag{7}$$

4.3 Poisson in h for the off-diagonal terms

Opening the two Kloosterman sums in (7), with $g = (c_1,c_2)$ and $q = [c_1,c_2] = c_1c_2/g$,

$$C(c_1,c_2) = \sideset{}{^*}\sum_{t_1 \bmod c_1}\ \sideset{}{^*}\sum_{t_2 \bmod c_2} e\Big(\frac{16Nt_1}{c_1}+\frac{16Nt_2}{c_2}\Big)\,\mathcal{G}(\lambda), \qquad \mathcal{G}(\lambda) = \sum_{h\in\mathbb{Z}^2} W\Big(\frac{h}{H}\Big) e\Big(\frac{\lambda|h|^2}{q}\Big), \quad \lambda = m\Big(\bar t_1\frac{c_2}{g}+\bar t_2\frac{c_1}{g}\Big).$$

The inner sum is a two-dimensional theta sum of length H and modulus q. Poisson summation (write $h = u + qv$, $u \bmod q$) gives, for $(\lambda,q) = 1$,

$$\mathcal{G}(\lambda) = \frac{H^2}{q^2}\sum_{\ell\in\mathbb{Z}^2} \hat W\Big(\frac{H\ell}{q}\Big) \sum_{u \bmod q} e\Big(\frac{\lambda|u|^2 + \ell\cdot u}{q}\Big) = \frac{H^2}{q^2}\sum_{\ell\in\mathbb{Z}^2} \hat W\Big(\frac{H\ell}{q}\Big) G(\lambda;q)^2\, e\Big(-\frac{\overline{4\lambda}\,|\ell|^2}{q}\Big),$$

with $|G(\lambda;q)|^2 = q$. The dual variable runs over $|\ell| \ll q/H$, so the dual sum has $(q/H)^2$ terms of size H^2/q, and the fixed $16N$ survives as the linear phase of the two-variable complete sum

$$\sum_{t_1,t_2} e\Big(\frac{16Nt_1}{c_1} + \frac{16Nt_2}{c_2} - \frac{\overline{4\lambda(t_1,t_2)}\,|\ell|^2}{q}\Big) G(\lambda(t_1,t_2);q)^2,$$

in which λ is a linear form in the inverses $\bar t_1, \bar t_2$: this is the "congruence equation in (t_1,t_2)" that the dualisation produces. Now the bookkeeping. The term $\ell = 0$ gives $(H^2/q)\chi(q)$ times a sum of $e(16Nt_1/c_1 + 16Nt_2/c_2)$ over (t_1,t_2) with $(\lambda,q) = 1$, a combination of Ramanujan sums, hence $O(H^2 g\, N^{\varepsilon}/q)$: the off-diagonal has no main term, in agreement with the random-sign behaviour of Part II, Section 8.2. For $\ell \ne 0$, even with square-root cancellation of Deligne type in the (t_1,t_2)-sum (which has $c_1c_2 = qg$ terms), the total is

$$\frac{H^2}{q^2}\cdot\Big(\frac{q}{H}\Big)^2 \cdot q \cdot (qg)^{1/2} = q^{3/2}g^{1/2},$$

against the trivial bound $H^2(c_1c_2)^{1/2} = H^2(qg)^{1/2}$ obtained by applying Weil to both Kloosterman sums before summing over h. The dualisation therefore loses the factor

$$\frac{q}{H^2} \ \ge\ \frac{c_1c_2/g}{mm'} \ \ge\ \frac{\sqrt{EN}}{m} \ \ge\ N^{1/2-\theta/4} \qquad \text{(balanced split } m \asymp m' \asymp N^{\theta/2}),$$

and it would gain only if $H > \sqrt{q}$, i.e. $mm' \gg \sqrt{EN}$, which never happens for $\theta < 2$. By (6) the side of the h-box is at most $\sqrt{mm'} = N^{\theta/2}$, the square root of the sieve level, while every modulus in (5) is at least $\sqrt{N/m}$; Poisson in h is counterproductive in the whole range of the sieve.

4.4 Conclusion, and what the data say

Neither Cauchy–Schwarz arrangement removes the fixed first argument: (A1) is weaker than the theta-function bound because it discards the saving on the k-sum, (A2) is a restatement, and the Poisson dual of the h-sum is longer than the sum itself by $N^{1/2-\theta/4}$. The only variable in (5) that is long compared with the modulus is k, and that is the modulus; its sum is exactly what the theta function already evaluates. A dispersion over the lattice points (x, y) themselves (the arrangement of Linnik's problem, the sum over m' inside the square) leads to the correlations $\sum_{m\alpha+[e_1,e_2]\beta=N} r(\alpha)r(e_1'\beta)r(e_2'\beta)$, whose weight $r(e_1'\beta)r(e_2'\beta)$ is not a theta coefficient and whose moduli $[e_1, e_2] \asymp E^2$ cost, with the theta bound, the level $1/3$; this is where the Deshouillers–Iwaniec machinery would have to enter, and it is the problem of Section 2.4, not a way around it.

The script `kloosterman_correlation.py` (`results/kloosterman_correlation.txt`) computes (7) for $N = 2000002$, $m = 5$, twelve primes $m' \in [41, 173]$ and $k = 43, 83, 127$, i.e. moduli $c \le 2.2 \cdot 10^4$ in the Kloosterman regime: there H is between 2.8 and 35, $H^2/c^2 = m/N = 2.5 \cdot 10^{-6}$ exactly, the diagonal $C(c, c)$ is 1.0–1.5 times the Weil size $H^2 c$, the off-diagonal correlations $C(c_1, c_2)/\sqrt{C(c_1, c_1)C(c_2, c_2)}$ have root mean square 0.38, 0.19, 0.14 (random signs on the $\asymp H^2$ lattice points would give 0.19, 0.10, 0.06), and the quadratic form $\sum_{m_1', m_2'} \pm \pm C$ is 0.65–1.02 times its trace for the sieve signs $\mu(m')$ and 0.95–0.99 for random signs, against 12 for full coherence: square-root cancellation over m', as in Part II, Section 8.2, and nothing that a dualisation of h could exploit.

5 Mollified theta weights and the dimension

Section 3.4 asked for remainder control for a sequence of dimension lower than 2. The weights

$$\tilde{w}(n) = r(n)\Big(\sum_{\substack{u \mid n \\ u \le N^{\xi}}} \gamma_u\Big)^2 r(N-n)\Big(\sum_{\substack{v \mid N-n \\ v \le N^{\xi}}} \gamma_v\Big)^2 \ge 0 \tag{8}$$

stay inside the theta framework: expanding the squares, $\sum_{d_1 \mid n,\, d_2 \mid N-n} \tilde{w}(n)$ is a combination of the sums $S([d_1, u_1, u_2], [d_2, v_1, v_2])$ with moduli $\le d_1 d_2 N^{4\xi}$, so that Part I, Theorem 3.2 controls the remainders of $\tilde{w}$ at the level $\theta_{\text{eff}} = 1/2 - 4\xi$ (two-sided mollifier) or $1/2 - 2\xi$ (mollifier on one side only). We determine exactly what such weights can and cannot do.

5.1 A mollifier below the sifting limit does nothing

Theorem 5.1. *Let (γ_u) be any weights supported on $u \le N^{\xi}$ with $\gamma_1 = 1$, let $\xi < a$ and $z = N^a$. Then $\tilde{w}(n) = w(n)$ for every n with $(n(N-n), P(z)) = 1$. Consequently $S(\tilde{\mathcal{A}}, z) = S(\mathcal{A}, z)$, $S(\tilde{\mathcal{A}}_p, z) = S(\mathcal{A}_p, z)$ for every $p \ge z$, and the weighted sum W of Section 3.1 is the same for $\tilde{w}$ and for w: the mollified sieve is the unmollified sieve, run at the level $1/2 - 4\xi$ instead of $1/2$.*

Proof. If $(n, P(z)) = 1$, the only divisor $u \le N^{\xi} < z$ of n is $u = 1$, so the inner sums in (8) are $\gamma_1 = 1$; the same for $N - n$. The sifted sums and the sums over $\mathcal{A}_p$, $p \ge z$, involve only such n. □

So for $\xi < a$ the answer to the question is unconditional: no weight of the form (8), non-negative or not, multiplicative or not, changes anything in the sieve except the level, which it lowers. The pairs (k, k') are read off from the thresholds at the level $1/2 - 4\xi$, which are worse than at $1/2$ (Part II, Figure 4(b)).

5.2 The density at a prime inside the mollifier

Proposition 5.2. *Let γ be multiplicative, supported on square-free integers, and ignore the truncation. The sequence $r(n)(\sum_{u|n}\gamma_u)^2$ has at a prime p the density*

$$\tilde{h}(p) = h(p)\,\frac{(1+\gamma_p)^2}{1+(2\gamma_p+\gamma_p^2)h(p)},$$

where h is the density of $r(n)$ (so $h(p)\sim 2/p$ for split p). In particular $\tilde{h}(p)=h(p)$ if $\gamma_p=0$, and $(1+\gamma_p)^2=\frac{1}{2}$ gives $\tilde{h}(p)=h(p)/(2-h(p))\sim 1/p$.

Proof. The main term of $\sum_{d|n} r(n)(\sum_{u|n}\gamma_u)^2$ is $X\sum_{u_1,u_2}\gamma_{u_1}\gamma_{u_2}h([d,u_1,u_2])$. For $d=1$ this is $X\prod_\ell(1+(2\gamma_\ell+\gamma_\ell^2)h(\ell))$, and for $d=p$ the Euler factor at p becomes $h(p)+(2\gamma_p+\gamma_p^2)h(p) = h(p)(1+\gamma_p)^2$, since $h([p,u_1,u_2]) = h(p)h([u_1,u_2])$ if $p\nmid u_1u_2$ and $= h([u_1,u_2])$ otherwise. Divide. □

Hence a mollifier changes the density exactly at the primes it contains, and only there: to lower the dimension at the primes of the Richert range $[N^a,N^b]$ it must contain them, i.e. have length at least N^b. The choice $(1+\gamma_p)^2=\frac{1}{2}$ for all p gives $(\sum_{u|n}\gamma_u)^2 = 2^{-\omega(n)}$ and $\tilde{w}(n) = 16\,b(n)b(N-n)$ on the square-free integers with split prime factors: the fully mollified weight is the indicator weight of Section 3.2, as it must be, and the truncation at N^ξ is precisely the trade between dimension and level.

5.3 The trade, quantitatively

With $\xi\ge a$ the sieve becomes a hybrid: the sifting range $p<z\le N^\xi$ has dimension $\kappa_0=1$ (two-sided) or $3/2$ (one-sided), the Richert range has density $\kappa(t)=\kappa_0$ for $t<\xi$ and 2 for $t>\xi$, and the level is $\theta_{\text{eff}} = 1/2-4\xi$ or $1/2-2\xi$. The weighted sieve of Section 3.1 then reads $L = f_{\kappa_0}(\theta_{\text{eff}}/a)$, $R=\int_a^b(1-t/b)\kappa(t)F_{\kappa_0}((\theta_{\text{eff}}-t)/a)\,\mathrm{d}t/t$, with the cut $b<\theta_{\text{eff}}$. Since the mollifier must reach the Richert range to act on it, $b<\theta_{\text{eff}}\le 1/2-4a$ (two-sided) or $1/2-2a$ (one-sided), which already forces $1/\lambda+2/b>2/b>4$ and leaves little room. The script `mollified_benchmark.py` scans (ξ,a,b) for both variants (the sieve constants for $\kappa=3/2$, $\alpha_{3/2}=3.9115$, $\beta_{3/2}=3.1158$, are computed by the solver of Part II with residual $5\cdot10^{-14}$; we have not cross-checked them against the published table) and finds (`results/mollified_benchmark.txt`):

	$\min(1/\lambda+2/b)$	$\min(1/\lambda+1/b)$	(k,k')
unmollified, $\kappa=2$, $\theta=1/2$ (Part II)	7.561	5.065	$(7,5)$
two-sided, $\kappa_0=1$, $\theta_{\text{eff}}=1/2-4\xi$	8.574	5.603	$(8,5)$
one-sided, $\kappa_0=3/2$, $\theta_{\text{eff}}=1/2-2\xi$	8.042	5.334	$(8,5)$
hypothetical: two-sided, no level cost	5.243	3.202	$(5,3)$

(The minima are attained at $\xi=a=0.035$, $b=0.355$ and $\xi=a=0.045$ in the two-sided case, at $\xi=a=0.0425$, $b=0.398$ and $\xi=a=0.05$ in the one-sided case, and at $\xi=0.44$, $b=0.49$ in the hypothetical one.) In both real variants the optimum sits at $\xi=a$, the smallest mollifier that touches the sifting range at all, and is worse than $(7,5)$: the level lost outweighs the dimension gained. The last line confirms Section 3: if the mollifier were free, pushing ξ to b would reproduce the $\kappa=1$ thresholds 5.18/3.14.

Remark 5.3. The conclusion is the precise form of the answer requested: (i) a mollifier truncated below N^a does not change the dimension at any prime $\ge N^a$, by Theorem 5.1, and does not change the sieve at all; (ii) to change the dimension at $p\in[N^a,N^b]$ the mollifier must contain p (Proposition 5.2), hence have length $\ge N^b$, hence cost $4b$ (or $2b$) in the level, which is more than the level itself for $b\ge 1/8$ (or $1/4$); (iii) the partial versions, mollifying only $[N^a,N^\xi]$,

lose more in level than they gain in dimension at every $\xi \geq a$. Selberg's upper-bound weights $(\sum_{u|n, u \leq U} \lambda_u)^2$ are of the same form and are subject to the same accounting. What is needed is a remainder term for $b(n)r(N-n)$ or $b(n)b(N-n)$ that does not pass through the moduli $[u_1, u_2]$, i.e. a non-modular input.

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